

L_RAG: A Controlled Ablation Study of Vietnamese Legal Retrieval-Augmented Generation

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Abstract

Legal question answering must retrieve evidence that is relevant, complete in its statutory context, valid at the relevant time, and traceable to an authoritative source. We evaluate L_RAG, a Vietnamese legal Retrieval-Augmented Generation pipeline that combines E5 dense retrieval, BM25, a knowledge-graph overlay, rank fusion, cross-encoder reranking, and citation-grounded generation. Across a 500-question benchmark, the ablations yield three practical findings. Adding metadata to embeddings improves document-level coverage, but not exact-chunk ranking. The tested hybrid and graph context-selection configurations yield lower exact-chunk coverage than DenseOnly, identifying opportunities in fusion and seed retention. Every tested cross-encoder improves ranking when candidate evidence is held fixed: Jina achieves the strongest retrieval scores, BGE the highest lexical answer-overlap scores, and mMiniLM the lowest reranking latency. A separate, fixed-context prompt comparison finds that GPT-4o-mini chain-of-thought prompting lowers lexical overlap, improves only template-based answerability diagnostics, and adds latency. Because the study has no human judgments of legal correctness or citation faithfulness, its conclusions are deliberately limited to retrieval, lexical overlap, abstention diagnostics, and recorded latency.

1 Introduction

Legal information systems must do more than return text that appears related to a question. A useful answer must be grounded in an authoritative provision, retain the qualifiers and exceptions that shape its meaning, and respect the provision’s temporal status. These requirements are especially demanding for Vietnamese legal corpora, where a citizen’s everyday wording may differ sharply from statutory language and where an answer can depend on a provision’s place in a legal hierarchy.

Retrieval-Augmented Generation (RAG) is a natural starting point: it gives a language model access to external evidence rather than relying solely on model parameters. However, a conventional dense-retrieval pipeline leaves several legal risks unresolved. It can match a query to a broad discussion rather than the governing clause; fixed-size chunks can separate a rule from its scope or exception; and semantic similarity alone does not determine whether a cited instrument remains valid. A legal RAG pipeline therefore needs to balance semantic matching, exact terminology, document structure, and evidence ordering before generation begins.

1.1 Problem Statement

This study examines four design choices intended to address those constraints:

1. **Vocabulary mismatch.** Users often describe a legal problem in everyday Vietnamese, whereas statutes use formal terms, identifiers, and numeric thresholds. Dense retrieval helps with paraphrase, while lexical retrieval can preserve exact legal terms.
2. **Lost statutory context.** Laws are hierarchical—from a legal instrument to chapters, articles, clauses, and items. A retrieved passage may be incomplete if its parent provision or an adjacent exception is missing.
3. **Validity and cross-reference.** Provisions can amend, replace, repeal, or cite one another. Retrieval should therefore account for authority, temporal status, and explicit legal relationships.

L_RAG combines dense and sparse retrieval with a graph overlay and a cross-encoder reranker. The central question is not whether every component is plausible in isolation, but which compo-

nents improve the evidence available to the generator under a controlled evaluation protocol.

1.2 Objectives

The project has four objectives:

- build an end-to-end Vietnamese legal RAG pipeline using dense E5 retrieval, BM25, reciprocal-rank fusion (RRF), reranking, and citation-grounded generation;
- represent legal hierarchy, reading order, and document relationships in a query-time knowledge-graph overlay;
- compare four multilingual rerankers under a shared candidate pool; and
- evaluate retrieval, answer-overlap, abstention, and latency behavior on a 500-question Vietnamese legal benchmark.

1.3 Contributions

This report contributes a controlled empirical account of L_RAG rather than an unqualified claim of legal reliability. Specifically, it provides:

1. a modular legal-RAG implementation that supports dense and sparse retrieval, graph-based context expansion, reranking, and cited generation;
2. a large Vietnamese legal corpus and a 500-question benchmark with chunk-, provision-, and document-level evidence provenance; and
3. four paired ablation families that distinguish document coverage from exact evidence ranking, isolate reranker effects at a fixed candidate pool, and separate observable prompt behavior from legal correctness.

1.4 Report Structure

Section 2 positions the study in RAG, legal IR, hybrid retrieval, and graph-augmented retrieval. Section 3 and Section 4 describe the system, corpus, and benchmark. Section 5 defines the comparisons and their boundaries. Section 6 and Section 7 report the results, while Section 8–9 interpret their practical implications and limitations.

2 Related Work

This section reviews existing literature and foundational paradigms relevant to Retrieval-Augmented Generation, domain-specific Legal Information Retrieval, hybrid retrieval strategies, and Knowledge-Graph-augmented systems.

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (Lewis et al., 2020) has emerged as a dominant framework to mitigate hallucinations, overcome outdated parameter knowledge, and ground Large Language Models (LLMs) in verifiable external knowledge. Standard RAG architectures operate in two main steps: (1) retrieving relevant document chunks from a vector store using dense bi-encoder embeddings, and (2) concatenating retrieved chunks into the prompt context for an LLM to generate an answer.

Despite their success in general-domain benchmarks, baseline RAG architectures face severe limitations in specialized, highly technical domains. In unstructured domain corpora, dense retrieval frequently suffers from low precision due to semantic ambiguity, while fixed-size text chunking loses structural dependencies spanning across paragraphs or document chapters. Advanced RAG paradigms—such as Self-RAG (Asai et al., 2024) and Corrective RAG (CRAG) (Yan et al., 2024)—introduce self-reflection and web-search fallback mechanisms, but they do not explicitly model formal domain taxonomies or statutory legal validity.

2.2 Legal Information Retrieval

Legal Information Retrieval (Legal IR) is a specialized subfield at the intersection of Information Retrieval and Legal Informatics (Locke and Zuccon, 2022). Legal IR differs fundamentally from open-domain text search due to four intrinsic characteristics of legal text:

1. **Hierarchical Structure:** Legal statutes are organized into strict logical trees (Code → Part → Chapter → Section → Article → Clause → Item).
2. **High Precision Requirements:** A single misplaced word or omission of an exception clause can alter the legal interpretation completely.
3. **Statutory Precedence:** Legal systems follow formal legal hierarchies (e.g., the Constitution overrides National Assembly Laws,

which in turn override Government Decrees and Ministry Circulars).

4. **Dynamic Validity & Amendments:** Legal provisions are constantly updated, amended, or repealed by subsequent legal instruments over time.

Early Legal IR research focused on rule-based citation parsing and expert systems. Recent approaches leverage specialized transformer models, such as Legal-BERT (Chalkidis et al., 2020) and Law-Former (Xiao et al., 2021), to encode statutory texts. However, relying solely on neural embeddings remains insufficient for capturing complex statutory relationships and temporal applicability.

2.3 Hybrid Retrieval and Reranking Fusion

To overcome the vocabulary mismatch problem of dense embeddings and the semantic rigidity of sparse keyword search, hybrid retrieval combines both modalities (Cormack et al., 2009). Sparse algorithms (e.g., BM25) excel at exact keyword matching—critical for capturing specific legal codes, numbers, and technical terminology—whereas dense bi-encoders (e.g., E5 (Wang et al., 2022)) excel at conceptual paraphrasing.

Combining retrieved lists from heterogeneous engines requires robust rank fusion. Reciprocal Rank Fusion (RRF) (Cormack et al., 2009) merges ranked outputs based on positional reciprocal scores:

$$\text{RRF}(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \quad (1)$$

where M is the set of retrieval systems, $r_m(d)$ is the rank of document d in system m , and k is a smoothing constant (typically $k = 60$).

Following initial retrieval, cross-encoder rerankers (Nogueira and Cho, 2019) evaluate full joint attention between the query and candidate passages (q, d) . Unlike bi-encoders, which compute separate vectors, cross-encoders achieve significantly higher ranking precision (measured via MRR and nDCG) at the expense of higher inference latency. Recent state-of-the-art multilingual rerankers include BGE-Reranker-v2-m3 (Chen et al., 2024), Jina-Reranker-v2 (Jina AI, 2024), and Qwen3-Reranker (Qwen Team, 2025).

2.4 Knowledge-Graph-Augmented Retrieval (GraphRAG)

Knowledge Graphs (KGs) represent domain entities and their explicit relationships as structured nodes and edges (e_1, r, e_2) . GraphRAG (Edge et al., 2024) integrates graph traversal with vector search, enabling LLMs to reason over multi-hop relationships and hierarchical structures.

In the legal domain, GraphRAG approaches construct statutory graphs linking documents, provisions, definitions, and legal concepts. At query time, traversal can expand seed vector hits through structural edges (HAS_PROVISION) and citation edges (CITES, AMENDS). This can restore statutory context, but its value depends on edge quality, the traversal policy, and the available context budget.

3 System Overview: L_RAG

This section details the end-to-end architecture of **L_RAG**, a Knowledge-Graph-augmented hybrid RAG system engineered for legal information retrieval and question answering over Vietnamese legal corpora.

3.1 Architecture

L_RAG connects corpus indexing, hybrid retrieval, graph-based context selection, candidate reranking, and cited generation. Figure 1 illustrates the end-to-end query pipeline across five core operational phases: (1) query scope and temporal constraint resolution, (2) hybrid dense-sparse retrieval (E5 + BM25), (3) knowledge-graph overlay filtering and context expansion, (4) rank fusion and candidate reranking, and (5) citation-grounded LLM prompting and answer generation. This modular overview shows the available components; each ablation activates the subset specified in Section 5.5. In particular, the retrieval/graph family evaluates structural expansion without reranking or query-time validity overlays.

3.2 Retrieval Components

The retrieval module combines three complementary retrieval approaches:

3.2.1 Dense Vector Search

Dense retrieval is powered by `intfloat/multilingual-e5-large`, a 560M parameter transformer embedding model. To solve the structural blindness of standalone chunks, each text chunk is preprocessed into a

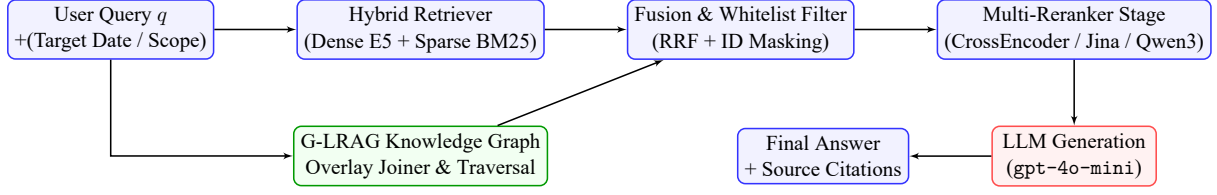


Figure 1: End-to-end operational workflow of the L_RAG system.

Metadata-Enriched Representation:

$$\text{Text}_{\text{embedded}} = \text{Doc_Title} \parallel \text{Provision_Header} \parallel \text{Chunk_Content} \quad (2)$$

Chunks are stored in an optimized FAISS vector database using IndexFlatIP with L2-normalized embeddings for fast cosine similarity search.

3.2.2 Sparse Keyword Search (BM25)

Sparse retrieval uses `rank_bm25` with Vietnamese word segmentation from `underthesea` and parameters $k_1 = 1.5$, $b = 0.75$. The retrieval/graph experiment partitions the corpus into 10 independently built BM25 shards to fit the available memory. Candidates from each shard are combined by BM25 score before dense-sparse RRF fusion. IDF and average document length are estimated within each shard; the results therefore characterize this practical sharded implementation. Shared corpus statistics and alternative shard-merging policies offer useful directions for refinement.

3.2.3 Knowledge Graph Overlay (G-LRAG Module)

The Knowledge Graph module acts as a query coordinator that generates dynamic document filter whitelists (`id_str_filter`) and performs horizontal reading-order expansions. Key features include:

- **Dynamic Overlay Folding:** Validity timelines are joined at query time relative to an `as_of_date` parameter to support filtering using the available validity metadata.
- **Precedence Sorting:** Candidates can be ordered using encoded legal-authority ranks:

$$\begin{aligned} \text{Hiến pháp} &\succ \text{Bộ luật / Luật} \\ &\succ \text{Nghị định} \succ \text{Thông tư} \end{aligned} \quad (3)$$

- **Reading-Order Traversal:** `CHUNK_NEXT` connects chunks within a provision, while

`PROVISION_NEXT` connects successive provisions. In the implemented hop convention, hop 1 covers the seed provision; additional hops reach subsequent provisions.

3.2.4 Fusion Layer

In the Hybrid retrieval/graph conditions, hit lists from dense search (R_{dense}) and sparse search (R_{BM25}) are merged using Reciprocal Rank Fusion (RRF) with constant $k = 60$. DenseOnly retains the dense ranking without a fusion stage.

3.3 Reranking Components

The reranker experiment selects $K = 30$ candidate chunks per query and scores the full (*query*, *chunk_text*) pair to refine the final ranking.

L_RAG supports four alternative reranking backends. Evaluation conditions load and run one backend at a time, rather than cascading all four per query:

1. **mMiniLMv2:** [cross-encoder/mmarco-mMiniLMv2-L12-H3](#) (Lightweight baseline, 117M params).
2. **BGE-Reranker-v2-m3:** [BAAI/bge-reranker-v2-m3](#) (Multilingual 567M params with sigmoid activation).
3. **Jina-Reranker-v2:** [jinaai/jina-reranker-v2-base-multilingual](#) (278M params, featuring dynamic position embedding patches).
4. **Qwen3-Reranker:** [Qwen/Qwen3-Reranker-0.6B](#) (CausalLM-based 0.6B parameter model evaluated in FP16 precision).

To prevent GPU Out-Of-Memory (OOM) failures during multi-model evaluation, the loader executes a strict **Sequential Load-Run-Unload Cycle**:

$$\begin{aligned} \text{Load}(M_i) &\longrightarrow \text{Execute}(M_i) \\ &\longrightarrow \text{Delete}(M_i) \longrightarrow \text{empty_cache}() \end{aligned} \quad (4)$$

3.4 Generation Component

The non-generator ablations supply up to $N = 10$ selected chunks to gpt-4o-mini at temperature $T = 0.0$. Context selection and output settings follow each experiment’s protocol. In the retrieval/graph family, the client sets no explicit output-token cap and uses a shared Vietnamese prompt with three evidence-oriented instructions:

1. **Evidence grounding:** Use only the supplied context and cite legal sources when present.
2. **Explicit abstention:** When evidence is insufficient, respond “*Không có đủ thông tin trong ngữ cảnh được cung cấp.*”
3. **Direct, scoped answers:** Answer naturally in Vietnamese, state any partial coverage, and omit internal reasoning. Benchmark answer types are evaluation labels, not prompt inputs in this family.

Source citations support inspection; semantic citation verification is outside the completed evaluation.

4 Dataset and Benchmark

This section describes the evidence base behind the experiments: a processed Vietnamese legal corpus, a graph that retains its structure and relationships, and a 500-question benchmark with explicit evidence provenance.

4.1 Corpus Description

The corpus contains Vietnamese legal instruments, including National Assembly Laws, Government Decrees, Prime Minister Decisions, and Ministry Circulars. The ingestion pipeline preserves document metadata, hierarchical provisions, and retrieval passages rather than treating each source as unstructured text. Table 1 summarizes the resulting scale.

Before indexing, each passage receives the metadata needed to reconnect it to its legal source:

Chunk_Meta = {chunk_id, doc_id, provision_id,
title, issuing_authority, effective_date} (5)

4.2 Knowledge-Graph Structure (G-LRAG Schema)

The G-LRAG graph makes statutory structure available at query time. It has three node types:

documents (D), provisions (P), and indexed chunks (C). Documents retain identifiers, authority rank, issuing body, effective date, and validity status; provisions retain their header and type; chunks retain their text and sequence number.

Two families of directed edges encode how evidence should be interpreted:

- **Structural edges** connect a document to its provisions, a provision to its chunks, adjacent chunks in reading order (CHUNK_NEXT), and adjacent provisions (PROVISION_NEXT). They support recovery of nearby statutory context.
- **Legal-relation edges** record amendments, replacements, and citations between legal instruments. They support validity-aware filtering and relationship-aware retrieval.

The graph is deliberately conservative. Draft or unverified records are quarantined at ingestion, and query-time traversal uses only edges whose direction has been verified. This protects retrieval from propagating a potentially incorrect legal relationship, although the remaining graph-quality limitations are discussed in Section 8.

4.3 QA Benchmark Dataset

The benchmark contains **500 Vietnamese legal questions**: 400 answerable questions and 100 deliberately unanswerable questions. Questions were created from bundles of approximately three to five related provisions and chunks using gemini-3.1-flash-lite. A separate verification pass checked whether a candidate was answerable from the supplied evidence, free of unsupported claims, and correctly labeled by category and difficulty. Only candidates that passed every check were retained.

The final generation run sampled 515 documents and 1,697 provisions, with a 72.41% verifier pass rate before diversity filtering and deduplication. The final set includes 265 single-hop, 106 citation, 96 legal-validity, 21 multi-hop, and 12 cross-document questions. Answer types span abstractive, boolean, extractive, and unanswerable responses. The unanswerable cases are constructed without supporting corpus evidence so that abstention behavior can be measured separately from answer generation.

Table 1: Processed Vietnamese legal corpus.

Entity Type	Description	Total Count
Documents	Full legal instruments (Laws, Decrees, etc.)	151,624
Provisions	Hierarchical units (Articles, Clauses, Items)	1,386,267
Chunks	Fixed/structural retrieval passages	1,513,376
Document Edges	Amend, replace, cite, and repeal relationships	883,256
External Stubs	Referenced external entities	19,763

4.4 Ground Truth Construction

Each query records its question, category, answer type, reference answer, and evidence provenance:

$$Q_i = \{\text{qa_id, question, category, answer_type, reference_answer, ground_truth}\}. \quad (6)$$

The ground-truth object identifies the supporting document, provision, and chunk IDs. Gold chunk IDs support Recall@ k , Hit@ k , MRR@ k , and nDCG@ k for answerable questions; unanswerable questions have empty evidence sets by design. This provenance makes the reported retrieval results auditable, but it does not replace expert judgment of whether an answer is legally complete or correct.

5 Experimental Setup

We evaluate L_RAG with controlled, paired ablations over the same Vietnamese legal QA benchmark. The benchmark contains 500 questions: 400 answerable and 100 unanswerable. Runs within an ablation family use the same benchmark, question identifiers, generator settings, and corpus/index version; only the component named by the family is changed. This design is important because the four completed families use different candidate budgets, as summarized in Table 2.

5.1 Research Questions

We organize the experiments around five questions:

1. Does adding document and provision meta-data to the embedding text improve exact evidence retrieval or document-level coverage?
2. Does hybrid dense-sparse retrieval improve coverage or ranking over dense retrieval alone?
3. Does the current G-LRAG structural context-expansion policy improve evidence coverage and downstream behavior for the tested legal QA benchmark?

4. Which reranker best trades ranking quality against inference cost when the candidate pool is held fixed?

5. Which generator reasoning strategy should be used? The completed generator-reasoning comparison contrasts base prompting with explicit chain-of-thought (CoT) prompting under a fixed retrieved context.

5.2 Metrics

For answerable questions, retrieval is evaluated against the annotated gold chunk IDs using Recall@ k , Hit@ k , MRR@ k , nDCG@ k , and Precision@ k . For graph runs we additionally report chunk-ID context precision, because a graph variant can return fewer chunks and therefore make a fixed-denominator precision score misleading. The reranker analysis reports candidate Recall@30 and candidate Hit@30 to distinguish the evidence ceiling from the ability of a reranker to promote evidence already present in the candidate pool.

For generated answers, we report exact match, token-level F1, and ROUGE-L on answerable questions. We also report the template-based answerability decision matrix (including non-refusal on answerable questions) and call its aggregate “decision accuracy” rather than treating it as a legal-safety metric. These automatic overlap and abstention measures are diagnostics rather than legal judgments: the available artifacts contain no human labels for legal correctness, completeness, claim faithfulness, or citation-to-chunk alignment. Those metrics are consequently not inferred from lexical overlap. For the retrieval/graph family, unanswerable accuracy is the fraction of the 100 unanswerable questions whose output contains the Vietnamese fallback phrase. It is distinct from the 500-case decision accuracy reported for the reranker and reasoning families.

Efficiency is reported as mean and median per-query latency, with p95 values where the source run recorded them. The reranker runs

used cached BM25 candidates and therefore report zero sparse-stage time; those values are not treated as full-online hybrid latency. In the retrieval/graph family, Hybrid timings include the BM25 batch-precomputation cost divided by 500 queries (26.990 seconds per query). These amortized timings describe computation accounting for the batch, rather than directly measured online request latency.

5.3 Evaluation Protocol

All completed runs use `intfloat/multilingual-e5-large` for dense retrieval and temperature 0.0. Output-token limits are family-specific: the retrieval/graph notebooks set no explicit client-side cap, whereas the generator-reasoning protocol uses a 1,024-token limit. The non-generator ablations use `gpt-4o-mini`; the reasoning family varies the generator model and prompt mode as listed in Table 2. The embedding-text ablation uses raw FAISS top-10 retrieval with no sparse, graph, fusion, or reranking module. The dense/hybrid/graph ablation retrieves up to 50 candidates per enabled retriever and selects a top-10 baseline context. Graph variants expand its top five seeds within their parent provisions (`max_hop=1`), replacing the baseline context with at most 10 expanded chunks. They use broad validity filtering, without query-time validity overlays or reranking. This comparison evaluates the complete context-selection policy, including seed selection and expansion. The reranker ablation reconstructs a shared RRF candidate pool of 30 chunks (C30) and compares the top 10 final chunks (T10). Its fusion constant is `rrf_k=60`.

Comparisons are paired by `qa_id`. Overall retrieval deltas use the 400 answerable questions. We report percentile bootstrap 95% confidence intervals for the mean per-question delta, using 10,000 resamples for the main comparisons and 2,000 resamples for the smaller reranker slices. A confidence interval containing zero is described as inconclusive rather than as evidence of an effect. The reranker candidate reconstruction agrees with the recorded RRF-only top-10 prefix for all 500 questions.

The generator-reasoning family holds the retrieval stack and intended retrieved top-10 context fixed for each question, and changes the prompt mode for the generator. The primary paired comparison uses the complete GPT-4o-mini

Base and CoT runs (500/500 questions, with 400 answerable pairs); their saved contexts match exactly. The DeepSeek-V3.1-Thinking, GLM-5, and Qwen3-8B pairs are also complete but remain descriptive cross-model checks. Four of the 500 shared DeepSeek cases have different exact top-10 chunk-ID sequences between Base and CoT. The Qwen Base and CoT manifests use different code snapshots, timeouts, and retry limits, so their latency and failure comparisons are descriptive. These conditions are not used for the primary paired claim. The post-hoc analysis and frozen result ledger are available at [ablation_study/e2e_LLM_reasoning/results/report.md](#) and [ablation_work/e2e_LLM_Reasoning/analyze_posthoc.py](#).

5.4 Evaluation Workflow and Derived Metrics

This subsection describes the generator-reasoning post-hoc analysis. Each condition is a saved 500-case run over shared question IDs, benchmark labels, and reference answers, with context and provenance alignment as described above. The post-hoc analysis reads these records and does not call a model or rerun retrieval. A row is valid when it is not marked failed, errored, or skipped, has no saved error, and contains a prediction. Lexical metrics use successful answerable rows; a paired Base/CoT statistic uses only IDs that satisfy these conditions in both runs and contain all three lexical metrics.

The reasoning-family evaluator normalizes text with Unicode NFC, lowercasing, punctuation-to-space replacement, and whitespace collapse. The derived metrics are separated below so that answer-quality outcomes are not conflated with retrieval, citation-structure, or operational diagnostics. The retrieval/graph notebooks instead lowercase text and collapse whitespace, retaining punctuation and using the complete answer for exact match. Lexical scores are therefore interpreted within each family using its saved evaluator.

Primary metrics. The primary generated-answer metrics are computed on successful answerable rows. For question i , let y_i be the normalized prediction and r_i the normalized reference. Exact Match (EM) is

$$EM_i = \begin{cases} 1, & y_i = r_i, \\ 0, & y_i \neq r_i, \end{cases} \quad EM = \frac{1}{N} \sum_{i=1}^N EM_i.$$

For boolean questions, the evaluator compares the first non-empty line. Let O_i be the multiset token overlap, L_i^y and L_i^r be the prediction and reference token counts, and C_i be the token-level longest common subsequence (LCS) length. Token F1 is

$$P_i = \frac{O_i}{L_i^y}, \quad R_i = \frac{O_i}{L_i^r}.$$

$$F_{1,i} = \frac{2P_i R_i}{P_i + R_i}.$$

ROUGE-L uses the same F-score form, replacing token overlap with the LCS:

$$P_i^L = \frac{C_i}{L_i^y}, \quad R_i^L = \frac{C_i}{L_i^r}.$$

$$\text{ROUGE-L}_i = \frac{2P_i^L R_i^L}{P_i^L + R_i^L}.$$

For a paired Base/CoT comparison, only question IDs with valid outputs and all three lexical metrics in both runs are retained. For any metric m , the per-question paired delta and its mean are

$$\Delta_i(m) = m_{\text{CoT},i} - m_{\text{Base},i}.$$

$$\bar{\Delta}(m) = \frac{1}{N_p} \sum_{i=1}^{N_p} \Delta_i(m).$$

Positive, zero, and negative deltas are counted as CoT wins, ties, and losses, respectively. Deltas are reported in percentage points for percentage metrics. Uncertainty is a percentile paired-bootstrap 95% confidence interval from 10,000 resamples (seed 42); EM additionally uses an exact McNemar test on discordant binary outcomes.

Additional diagnostics. These measures describe evidence availability, answer/refuse behavior, citation structure, reliability, and efficiency. They are not direct tests of legal correctness, citation entailment, or hidden reasoning.

Context Recall@ k . Let G_i be the gold chunk-ID set and $R_{i,k}$ the retrieved top- k set:

$$\text{ContextRecall}@k_i = \frac{|G_i \cap R_{i,k}|}{|G_i|}.$$

Answerable cases are stratified as fully retrieved, partially retrieved, or absent according to whether this value is one, between zero and one, or zero.

Decision accuracy. The persisted `unanswerable_accuracy` detector is mapped to answer/refuse decisions. An answerable refusal is

incorrect, while a refusal on an unanswerable question is correct. The two reported denominators are

$$\text{DecisionAccuracy}_{\text{scheduled}} = \frac{C_{\text{scheduled}}}{500},$$

$$\text{DecisionAccuracy}_{\text{valid}} = \frac{C_{\text{valid}}}{N_{\text{valid}}}.$$

Here $C_{\text{scheduled}}$ is the number of correct decisions among all 500 scheduled rows, C_{valid} is the number among valid rows, and N_{valid} is the number of valid rows.

Citation and structural diagnostics. Citation presence is the fraction of successful answerable answers containing at least one citation marker:

$$\text{CitationPresence} = \frac{A_{\text{marker}}}{A_{\text{successful}}}.$$

Here A_{marker} is the number of successful answerable answers with at least one marker and $A_{\text{successful}}$ is the number of successful answerable answers. If F_i^{marked} is the number of scored factual sentences with a marker and F_i^{scored} is the total number of scored factual sentences, structural coverage is

$$\text{StructuralCoverage} = \frac{1}{N} \sum_{i=1}^N \frac{F_i^{\text{marked}}}{F_i^{\text{scored}}}.$$

Invalid-marker rate is the fraction of all markers that are invalid:

$$\text{InvalidMarkerRate} = \frac{M_{\text{invalid}}}{M_{\text{all}}}.$$

Completion and latency. Completion rate is

$$\text{CompletionRate} = \frac{S}{Q}.$$

Here S is the number of successful rows and Q is the number of scheduled rows. For each saved latency summary, conversion from milliseconds to seconds is

$$\ell_{\text{seconds}} = \frac{\ell_{\text{milliseconds}}}{1000}.$$

The report uses the saved mean, median (p50), and p95 values. Reranker runs used cached BM25 candidates, so their timings are not full-online hybrid latency.

5.5 Ablation Configurations

Table 2 lists the frozen configurations used in the completed analyses. Names are aligned with the run manifests and the derived reports under `ablation_work/`.

Table 2: Completed ablation configurations. A dash means that the component was disabled. Candidate budgets are shown because they differ by ablation family.

Family	Configuration	Dense text	Sparse	Graph	Reranking	Candidate/final
Embedding	Chunk-only	chunk text	—	—	—	10/10
Embedding	Chunk+metadata	header + chunk	—	—	—	10/10
Retrieval/graph	DenseOnly	header + chunk	—	—	—	50/10
Retrieval/graph	HybridOnly	header + chunk	BM25	—	—	50/10
Retrieval/graph	DenseGraph	header + chunk	—	G-LRAG	—	50/10
Retrieval/graph	HybridGraph	header + chunk	BM25	G-LRAG	—	50/10
Reranker	None	shared RRF pool	BM25	—	none	30/10
Reranker	RRF-only	shared RRF pool	BM25	—	RRF	30/10
Reranker	mMiniLM	shared RRF pool	BM25	—	mMiniLM CE	30/10
Reranker	Qwen3	shared RRF pool	BM25	—	Qwen3	30/10
Reranker	BGE	shared RRF pool	BM25	—	BGE-v2-m3	30/10
Reranker	Jina	shared RRF pool	BM25	—	Jina-v2	30/10
LLM reasoning	GPT-4o-mini Base	fixed context	—	—	Base prompt	fixed/10
LLM reasoning	GPT-4o-mini CoT	fixed context	—	—	CoT prompt	fixed/10

5.6 Reproducibility and Artifact Ledger

Each run is accompanied by a manifest, per-question outputs, metric summary, and latency file. The derived analyses are versioned alongside the source outputs: the embedding comparison is in [ablation_study/chunkOnly_vs_chunkMetadata/results/](#), the dense/hybrid/graph comparison is in [ablation_study/dense_vs_Hybrid_vs_Graph/results/](#), and the multi-reranker analysis is in [ablation_work/reranker_choices/evaluation_runs/ablation3_reranker_v2/](#). The script that reconstructs the reranker candidate pool and produces its paired analysis is [ablation_work/reranker_choices/analyze_reranker_metrics.py](#).

6 Ablation Study

The ablations answer a practical design question: which additions improve the evidence that reaches the generator, and which merely add cost or complexity? We evaluate four families: embedding text, retrieval and graph expansion, reranking, and generator prompting. Candidate construction differs across families, so comparisons are made only within a family; the study is not a single pooled factorial experiment.

6.1 Embedding Text: Chunk-Only versus Chunk+Metadata

This ablation changes only the text indexed by the dense encoder: either the chunk body or a header-plus-chunk representation. The encoder, generator, benchmark, seed, and raw FAISS evaluation path are unchanged. On 400 paired answerable

questions, metadata leaves Recall@10 unchanged (48.5% for both conditions) but raises document-level success from 65.8% to 70.2%. The paired gain in document success is +4.5 percentage points (pp; 95% CI [+0.5,+8.8]), while MRR@10 falls by 5.9 pp (95% CI [−9.1, −2.7]).

The result separates two levels of retrieval quality. Metadata makes it more likely that the system enters the correct legal instrument, but it does not promote the exact supporting chunk earlier in the ranking. It should therefore be treated as a document-discovery aid, not as an unconditional improvement in provision-level retrieval. Table 4 provides the complete comparison.

6.2 Retrieval Mode and Graph Expansion

The second family compares DenseOnly, HybridOnly, DenseGraph, and HybridGraph. Hybrid retrieval adds BM25 and RRF to the same dense index; graph variants expand the top five seeds within their parent provisions and replace the baseline context with at most 10 expanded chunks. The comparison evaluates this complete context-selection policy. DenseOnly is the strongest configuration in this family: it exceeds HybridOnly on Recall@10 (48.55% vs. 42.57%) and nDCG@10 (0.2813 vs. 0.2638). The paired Recall@10 change for HybridOnly is −5.98 pp (95% CI [−9.71, −2.21]).

The current graph policy also reduces coverage. Relative to DenseOnly, DenseGraph lowers Recall@10 by 12.40 pp; it removes at least one previously retrieved gold chunk for 61 questions and adds a graph-only gold chunk for only one. Relative to HybridOnly, HybridGraph lowers Re-

call@10 by 11.81 pp, with 59 questions losing gold evidence and six gaining it. Graph contexts are more concentrated—5.605 chunks on average for DenseGraph and 6.305 for HybridGraph over the 400 answerable questions—but cover fewer gold chunks. This identifies seed retention as a concrete opportunity to improve coverage while retaining local structural context.

6.3 Reranker Choice at a Fixed Candidate Pool

The third family provides the cleanest component comparison. It evaluates no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina over the same reconstructed RRF pool of 30 candidates and retains the top 10. All six runs cover the same 500 questions with no retrieval or generation errors, and the reconstruction matches the recorded RRF-only top-10 prefix for all 500 cases. Rerankers jointly score each query–candidate pair, capturing richer cross-attention interactions than comparing separately encoded query and document vectors. Candidate Recall@30 (0.5995) and Candidate Hit@30 (0.6450) are identical across conditions, so the observed differences arise entirely from re-ordering and retaining the same available evidence.

Jina is the strongest overall retrieval configuration. Relative to RRF-only, it improves Recall@10 by +9.23 pp, MRR@10 by +13.35 pp, and nDCG@10 by +11.85 pp; all paired 95% bootstrap confidence intervals exclude zero. Qwen3 and BGE also show strong positive retrieval effects, while mMiniLM delivers a smaller but reliable gain at the lowest reranking latency (0.385 s vs. 9.695 s for Qwen3). BGE achieves the highest observed Token F1 (0.2955) and ROUGE-L (0.2603), while Jina achieves the highest recorded unanswerable accuracy (0.884). These lexical and template-based diagnostics reflect output structure rather than formal legal correctness. Table 6 and Table 7 provide the aggregate and paired results.

6.4 LLM Generator Ablation: Base versus CoT

This ablation asks whether a reasoning-oriented prompt changes observable answer behavior when the model receives the same legal material. Every condition uses the 500-question benchmark (400 answerable and 100 unanswerable questions), the same top-10 context, the dense-graph-RRF-mMiniLM retrieval stack, temperature 0, and a

1,024-token output limit. GPT-4o-mini Base versus CoT is the pre-specified primary comparison: both runs complete all 500 cases and have identical ordered top-10 chunk-ID sequences for every question.

The analysis reads frozen predictions, metrics, latency records, error files, and manifests; it does not rerun retrieval or generation. Deltas are CoT minus Base, with 95% percentile confidence intervals from 10,000 paired bootstrap resamples (seed 42). Exact Match also uses an exact McNemar test. Section 5.4 defines the metrics and denominators.

6.4.1 Primary paired result

For GPT-4o-mini, CoT lowers Token F1 from 24.72% to 23.15% (−1.57 pp; 95% CI [−2.24, −0.92]) and ROUGE-L from 21.60% to 19.90% (−1.71 pp; 95% CI [−2.34, −1.10]). Both intervals exclude zero. Exact Match changes by only −0.25 pp, with an interval that includes zero (McNemar $p = 1.00$). CoT wins, ties, and loses on Token F1 in 79, 164, and 157 cases, respectively: the decline is broad but not universal. It also adds 0.76 seconds to median generation and end-to-end latency, and 1.21 seconds to end-to-end p95 latency.

6.4.2 Behavior conditional on available evidence

Because retrieval is fixed, Recall, MRR, and nDCG are controls rather than LLM outcomes. We instead group answerable cases by saved context Recall@10. CoT lowers Token F1 when all gold support is present (−1.98 pp; $n = 138$), when support is partial (−1.86 pp; $n = 29$), and when no gold support is retrieved (−1.30 pp; $n = 233$). ROUGE-L changes in the same direction (−2.30, −1.59, and −1.37 pp). Exact Match rises by 0.72 pp only in the fully retrieved group; this small descriptive change does not reverse the primary result. Evidence availability describes the context, not whether the answer is faithful to it.

6.4.3 Answerability decisions

The saved field historically called `unanswerable_accuracy` is a template-based answer/refuse check over both answerable and unanswerable questions. It is not a direct measure of refusal quality. Both GPT prompt modes refuse 98 of the 100 unanswerable cases and answer the remaining two. CoT’s 1.40-pp gain in overall decision accuracy comes entirely from

answering seven more answerable cases, reducing template-detected false refusals from 85 to 78. The artifacts do not preserve a separate clarification decision, so the result cannot establish whether the additional answers were appropriate.

6.4.4 Citation structure, reliability, and descriptive model checks

Table 3 consolidates all four models and both prompt modes. For the primary GPT pair, CoT raises citation presence from 78.25% to 81.00% and mean unique sources from 1.46 to 1.68, while structural citation coverage falls from 89.42% to 87.91%. Both runs have zero invalid markers. These are formatting and placement diagnostics only; they do not show that a cited passage entails the adjacent legal claim.

The other model pairs are descriptive checks, not additional primary tests. Their CoT-minus-Base Token-F1 changes are -1.33 pp for DeepSeek-V3.1-Thinking (95% CI $[-1.96, -0.73]$), -0.68 pp for GLM-5 ($[-1.23, -0.11]$), and -1.92 pp for Qwen3-8B ($[-2.77, -1.15]$). Four DeepSeek cases have different ordered top-10 chunk sequences across prompts. The Qwen manifests also differ in code snapshot, timeout, and retry limits, so its latency and failure comparison is not strictly controlled. No prompt effect is pooled across models.

Claim boundary. The saved artifacts contain final answers, lexical scores, retrieved contexts, structural citation fields, and latency records. They do not contain legal correctness labels, atomic-claim support labels, citation-entailment judgments, preferred answer/clarify/refuse labels, raw chain-of-thought, or reasoning-token accounting. The primary ablation therefore supports a narrow conclusion: GPT-4o-mini CoT reduces template-detected false refusals and increases citation presence, but lowers lexical overlap across all evidence-availability strata and adds latency. It does not establish better or worse reasoning, legal correctness, semantic grounding, citation entailment, or refusal quality.

7 Results

The results tell a consistent story: better candidate ordering is more useful than adding retrieval stages under the current settings. The tables below separate document discovery, exact evidence ranking, answer-overlap diagnostics, and latency so that a

gain in one dimension is not mistaken for a general quality improvement. The generator-reasoning ablation is reported separately in Section 6.4. Retrieval metrics use answerable questions and are paired within each experiment family. Deltas are reported in percentage points (pp), not relative percentages.

7.1 Embedding Representation

Adding metadata changes the level at which retrieval succeeds. At top 10, exact chunk recall is unchanged to one decimal place, while document success increases by 4.5 pp. At the same time, the paired MRR decrease is reliably below zero. Metadata therefore helps the system locate the right legal instrument, but it does not improve—and may delay—the first exact supporting chunk.

7.2 Dense, Hybrid, and Graph Retrieval

DenseOnly is the strongest exact-chunk retriever in this family. HybridOnly is lower on Recall@10 and nDCG@10, and its paired Recall loss has a 95% interval entirely below zero. Graph expansion lowers Recall@10 for both seed retrievers: by 12.40 pp from DenseOnly and 11.81 pp from HybridOnly. The per-query evidence counts explain the decline: expansion adds a gold chunk for only one DenseOnly query and six HybridOnly queries, while removing at least one previously retrieved gold chunk for 61 and 59 queries, respectively.

Graph contexts are denser: chunk-ID precision rises from 0.0580 to 0.0800 for Dense and from 0.0493 to 0.0641 for Hybrid. That gain must be read beside the much larger recall loss. Fall-back detection on the 100 unanswerable questions also rises (0.38 to 0.41 for Dense and 0.38 to 0.47 for Hybrid), but paired Token F1 changes are small and inconclusive. The present graph policy thus creates a precision–coverage trade-off, not a demonstrated overall improvement.

7.3 Reranker Comparison

The fixed C30 pool makes this the most interpretable comparison: every reranker sees the same candidate evidence. Jina achieves the strongest retrieval scores, BGE the highest lexical-overlap scores, and Qwen3 competitive quality at a much larger reranker-stage cost. The recorded unanswerable column is a template-based diagnostic, not semantic legal-answer evaluation.

All rerankers have positive paired intervals for the three primary ranking measures, whereas ev-

Table 3: Generator-prompt ablation. EM, Token F1, ROUGE-L, and citation metrics use 400 answerable outputs; template-based decision accuracy uses all 500 cases. Latencies are p50/p95 seconds. Structural coverage measures citation-marker placement, not semantic entailment.

Model	Prompt	EM	F1	R-L	Decision	Cite pres.	Struct. cov.	Sources	Gen. p50/p95	Total p50/p95
GPT-4o-mini	Base	14.00	24.72	21.60	82.60%	78.25%	89.42%	1.46	2.92/5.74	13.99/17.21
GPT-4o-mini	CoT	13.75	23.15	19.90	84.00%	81.00%	87.91%	1.68	3.68/6.17	14.74/18.42
DeepSeek-V3.1-Thinking	Base	22.25	17.84	15.28	92.00%	90.00%	82.38%	2.57	7.28/15.51	16.15/24.55
DeepSeek-V3.1-Thinking	CoT	22.25	16.51	14.12	92.20%	90.75%	75.21%	2.71	7.82/21.02	18.52/31.37
GLM-5	Base	25.25	16.91	14.90	91.80%	91.75%	80.26%	2.72	4.17/8.79	14.29/19.09
GLM-5	CoT	25.00	16.23	14.27	93.40%	93.75%	78.12%	3.05	4.15/8.73	15.96/21.23
Qwen3-8B	Base	19.25	21.98	18.92	95.80%	97.75%	65.78%	1.79	4.77/9.13	14.15/18.51
Qwen3-8B	CoT	17.25	20.05	17.17	95.20%	98.25%	58.15%	1.81	4.97/10.47	14.58/20.16

Table 4: Embedding-text ablation on 400 paired answerable questions. Values are percentages; “document success” means that a retrieved chunk from the gold document is present in the indicated context. Deltas are chunk+metadata minus chunk-only.

Variant	Hit@5	Recall@5	MRR@5	Doc.@5	Hit@10	Recall@10	MRR@10	Doc.@10
Chunk-only	44.0	39.0	27.4	59.0	53.5	48.5	28.6	65.8
Chunk+metadata	40.2	36.8	20.9	61.8	53.2	48.5	22.7	70.2
Delta	−3.8	−2.2	−6.5	+2.8	−0.2	+0.1	−5.9	+4.5

ery false-refusal interval includes zero. Jina retains 0.8850 of the gold chunks available in C30, compared with 0.7129 for RRF-only. Its advantage therefore reflects better promotion of available evidence, not a larger candidate pool. The full safety and win/tie/loss analysis is stored with the derived artifacts.

7.4 Category and Answer-Type Analysis

The largest Jina gains occur for legal-validity and citation questions, while the smaller extractive slice is inconclusive. Table 8 reports the requested Jina-versus-RRF comparison; all slices are answerable cases, and intervals use 2,000 paired bootstrap resamples. The multi-hop slice has only 20 questions and is exploratory. The hard difficulty and cross-document slices are also too small to support a main conclusion.

7.5 Generation and Efficiency

Automatic answer metrics do not establish legal correctness. Metadata changes Token F1 by +0.1 pp and ROUGE-L by −0.1 pp; graph expansion changes Token F1 by approximately +0.4 pp for Dense and +0.2 pp for Hybrid, with bootstrap intervals crossing zero. In the reranker family, BGE has the strongest observed overlap scores, while Jina has the best retrieval scores and a lower total p50 than BGE (4.158 s vs. 5.848 s). These results support a conditional engineering choice, not a claim that either reranker generates more legally correct answers.

The batch accounting makes the sparse-stage cost visible. Hybrid total median time is 31.538 s, compared with 5.483 s for DenseOnly, and includes 26.990 s per query allocated from 13,495 s of BM25 precomputation over the 500-question batch. The batch cost is already included in these per-query values and is distinct from index construction. These timings guide resource planning for the evaluated implementation; online latency requires a separate measurement. Among rerankers, mMiniLM is fastest (0.385 s reranker-stage p50), Jina is intermediate (1.278 s), BGE is slower (2.872 s), and Qwen3 is much slower (9.695 s). The reranker artifacts do not record p95, peak GPU memory, or full-online sparse latency, so those deployment quantities remain open.

8 Discussion

The ablations replace a broad “more components are better” assumption with a more selective conclusion. In the current benchmark and implementation, components that improve document context do not necessarily improve exact evidence retrieval, and components that are structurally appealing can still harm coverage when the context budget is fixed. The strongest result is not a larger retrieval stack; it is better ranking of a fixed candidate pool.

8.1 What the Findings Mean

Metadata changes the level of success. Adding document and provision metadata helps the re-

Table 5: Dense/hybrid retrieval and graph results. Retrieval values use 400 answerable questions, including average context size. “Chunk-ID precision” is the per-query fraction of returned chunks matching gold chunk IDs. p50/p95 timings use all 500 questions. Hybrid timings include 26.990 seconds per query of amortized BM25 batch computation, rather than measured online BM25 request latency.

Method	Recall@10	MRR@10	nDCG@10	Chunk-ID precision	Avg. chunks	Retrieval p50/p95	Total p50/p95
DenseOnly	0.4855	0.2267	0.2813	0.0580	10.000	0.797/0.856	5.483/9.946
HybridOnly	0.4257	0.2252	0.2638	0.0493	10.000	27.855/27.924	31.538/33.623
DenseGraph	0.3615	0.2045	0.2374	0.0800	5.605	0.798/0.860	4.998/9.750
HybridGraph	0.3076	0.2012	0.2228	0.0641	6.305	27.858/27.929	31.180/33.312

Table 6: Multi-reranker ablation over a shared 30-candidate pool and final top-10 context. Retrieval and overlap values are means over 400 answerable questions; recorded unanswerable accuracy and p50 latency use the complete 500-case runs.

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Rec. unans. acc.	Rerank p50	Total p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	0.856	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	0.874	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	0.872	0.385	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	0.876	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	0.2955	0.2603	0.868	2.872	5.848
Jina	0.5305	0.5850	0.3670	0.3914	0.2913	0.2558	0.884	1.278	4.158

triever identify the right legal instrument, but does not improve the rank of the first exact gold chunk. The header tokens likely provide useful document-level cues while competing with passage content when the target is a specific provision. The practical implication is to evaluate both document coverage and chunk ranking: reporting only one would hide this trade-off.

Hybrid and graph results identify targeted refinements. DenseOnly provides the strongest exact-chunk coverage in this family. The Hybrid result motivates examining tokenization, shard-local BM25 statistics, fusion, and candidate allocation. Its reported timings include amortized batch computation, so online serving efficiency remains a separate engineering question.

Structural expansion produces denser contexts, with a coverage trade-off under the current replacement policy. It starts from five seeds, whereas the baseline retains ten chunks. Of the 65 gold-chunk occurrences lost by DenseGraph, 60 were outside the original top five; the corresponding count for HybridGraph is 52 of 60. These observations motivate preserving the baseline evidence before adding structural context. The comparison evaluates seed selection, ordering, and expansion together; it does not isolate their individual effects or test the full validity-overlay functionality.

Reranking is the clearest robust improvement. With candidate evidence held fixed, every evaluated cross-encoder improves Recall@10,

MRR@10, and nDCG@10 over RRF-only. Jina has the largest paired gains and retains substantially more of the available gold evidence, showing that its advantage comes from promotion rather than access to a larger pool. This is strong retrieval evidence, although it is still not evidence that the generator’s legal claims are correct.

Reasoning prompts change output behavior, not a validated legal quality measure. For the controlled GPT-4o-mini comparison, CoT lowers lexical overlap across all evidence-availability strata and increases latency. It reduces template-detected false refusals and increases citation-marker presence, but the study does not contain human evaluations of helpfulness, correctness, or citation entailment. The defensible conclusion is therefore operational: this CoT prompt has formatting-related benefits but no demonstrated legal-answer quality advantage.

8.2 Design Implications

The findings motivate evaluating dense retrieval followed by a cross-encoder as a streamlined deployment candidate. Jina is a promising choice based on its ranking and latency results in the tested dense-BM25 RRF pool; its performance on a dense-only pool remains to be measured. Within the evaluated pool, mMiniLM offers the lowest reranking latency and BGE the highest observed lexical overlap. These results guide the next configuration comparison.

Further Hybrid evaluation should measure on-

Table 7: Paired effects of each reranker relative to RRF-only. Deltas are in percentage points; intervals are 95% percentile bootstrap CIs from 10,000 question-level resamples over the 400 answerable cases. A negative false-refusal delta indicates fewer refusals on answerable questions.

Reranker	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Δ false refusal
mMiniLM	+5.91 [+1.85, +9.96]	+7.12 [+3.55, +10.68]	+6.96 [+3.77, +10.26]	+0.25 [−3.00, +3.50]
Qwen3	+8.37 [+4.57, +12.25]	+10.84 [+7.24, +14.36]	+9.60 [+6.52, +12.73]	−0.25 [−3.25, +2.75]
BGE	+8.07 [+4.29, +11.80]	+9.12 [+5.50, +12.70]	+8.65 [+5.47, +11.84]	+0.75 [−2.25, +3.75]
Jina	+9.23 [+5.51, +12.98]	+13.35 [+9.51, +17.24]	+11.85 [+8.46, +15.27]	−1.25 [−4.00, +1.25]

Table 8: Jina versus RRF-only on selected category and answer-type slices. Deltas are percentage points with 95% paired-bootstrap CIs.

Dimension	Slice	<i>n</i>	Δ Recall@10	Δ MRR@10	Δ nDCG@10
Category	citation	106	+11.89 [+2.82, +21.70]	+16.22 [+7.86, +24.51]	+15.82 [+7.51, +23.82]
Category	legal validity	91	+14.47 [+6.96, +21.43]	+18.90 [+10.28, +27.26]	+15.22 [+9.13, +21.50]
Category	single hop	172	+5.04 [−0.00, +10.37]	+9.63 [+4.34, +14.69]	+8.41 [+3.81, +13.05]
Category	multi hop	20	+7.50 [−0.83, +17.50]	+10.12 [−3.42, +24.33]	+7.95 [−0.77, +17.98]
Answer type	extractive	78	+0.13 [−8.85, +9.74]	+8.53 [−0.90, +17.39]	+7.14 [−1.33, +16.23]
Answer type	abstractive	199	+13.65 [+8.50, +18.80]	+14.74 [+9.61, +20.14]	+13.81 [+9.56, +18.71]
Answer type	boolean	123	+7.86 [+1.63, +14.50]	+14.18 [+7.64, +21.13]	+11.68 [+6.27, +17.95]

line BM25 cost and examine shared corpus statistics, tokenization, fusion, and candidate allocation. Graph augmentation should be introduced as a targeted, evidence-preserving mode for validity or relationship-heavy questions, rather than as unconditional expansion. A useful next graph policy would reserve space for high-confidence seed chunks before adding adjacent or related context.

Regardless of configuration, a production legal system must surface source citations and validity metadata, abstain when evidence is insufficient, and direct users to verify the cited instrument’s current status. None of the automatic metrics in this study can certify a legal answer’s effect or safety.

8.3 Threats to Validity

The benchmark has 500 questions, with relatively small multi-hop, hard, and cross-document slices. Its gold chunk sets may omit passages that a legal expert would regard as sufficient, and document-level success is not the same as provision-level correctness. Each ablation family uses one embedding model and one corpus/index version, but candidate budgets differ across families; absolute scores should therefore not be pooled across them.

The reranker and reasoning safety analyses rely on deterministic refusal-marker and template classifiers rather than human assessment. The reasoning artifacts lack raw CoT, token counts, cost records, semantic claim labels, and citation-entailment judgments. Four DeepSeek cases also have different saved context orders across prompts, and the Qwen Base/CoT manifests differ in code

snapshot, timeout, and retry configuration. Those comparisons remain descriptive.

Finally, cached BM25 candidates mean that reranker latency is not full-online hybrid latency; p95 latency, peak GPU memory, and controlled throughput are not available for every configuration. The graph corpus also has unresolved relationship-direction and validity-annotation questions. We therefore frame the results as benchmark-bounded engineering evidence, not a claim of legal reliability or deployment readiness.

Limitations

The corpus can contain missing body text, incomplete metadata, external references, and uncertain relationship direction. The in-memory graph also has a substantial memory footprint at the current corpus scale. More importantly, retrieval and generation results are bounded by the benchmark and its evidence annotations: a fluent answer, lexical overlap, or well-formed citation is not evidence of legal correctness.

The prompt analysis is limited to saved output records. It contains no raw chain-of-thought, reasoning-token counts, cost data, semantic claim labels, or human assessment of citation entailment. Four DeepSeek Base/CoT cases have different ordered top-10 chunk sequences, and the Qwen3-8B Base/CoT manifests use different code snapshots, timeouts, and retry limits. Those model-pair comparisons are consequently descriptive. L_RAG is a

research prototype and must not substitute for qualified legal review.

Ethical Considerations

Legal QA may affect access to rights, benefits, obligations, and remedies. Systems built on this work should show the cited source and validity metadata, make abstention visible, and encourage users to verify the current legal instrument before acting. Evaluation and deployment must also account for uneven corpus coverage, language variation, privacy in user queries, and the risk that a confident automated explanation receives undue trust.

9 Conclusion

This controlled ablation study identifies effective components and concrete directions for refining L_RAG. Metadata-enriched embeddings improve document-level coverage but not exact-chunk ranking. DenseOnly leads coverage in the retrieval/graph family, while the sharded Hybrid implementation and top-five-seed graph replacement policy highlight opportunities in fusion, serving efficiency, and evidence retention. With the candidate pool fixed, cross-encoder reranking provides consistent gains: Jina improves Recall@10, MRR@10, and nDCG@10 over RRF-only with paired confidence intervals above zero. BGE leads the observed lexical overlap metrics, while mMiniLM has the lowest reranking cost.

The fixed-context generator ablation adds an important qualification. For GPT-4o-mini, CoT prompting lowers Token F1 by 1.57 pp and ROUGE-L by 1.71 pp, improves only template-based answerability diagnostics, and increases latency. It is therefore not a demonstrated improvement in legal-answer quality.

Taken together, the results motivate testing a dense-retrieval-plus-reranker baseline. Jina is a promising candidate based on the evaluated RRF pool; transferring that advantage to a dense-only pool requires a direct comparison. The findings do not establish legal correctness, citation faithfulness, or safe autonomous legal advice: the available artifacts lack human semantic evaluations.

9.1 Future Work

The next phase should add blinded expert labels for legal correctness, completeness, claim faithfulness, citation precision, and citation recall.

It should also measure full-online p95 latency, GPU memory, and throughput; tune candidate-pool size, RRF, and graph budget; and validate graph relation direction against authoritative document pairs. The most promising technical change is seed-preserving graph expansion, which can add surrounding context without discarding high-confidence retrieved evidence. Future prompt studies should rerun all conditions with provenance-aligned manifests and evaluate answers, citations, and abstentions semantically rather than through lexical or template-based proxies.

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A Dataset and Graph Schema

The corpus represents legal instruments as documents, hierarchical provisions, and indexed chunks. A document stores its identifier, title, issuing authority, effective-date and validity metadata; a provision stores its article/clause hierarchy; and a chunk stores passage text, parent identifiers, and reading order. The graph uses containment edges `DOCUMENT_HAS_PROVISION` and `PROVISION_HAS_CHUNK`, reading-order edges `CHUNK_NEXT` and `PROVISION_NEXT`, and document-level legal relationships such as amendment, replacement, repeal, and citation. The completed evaluation uses the chunk identifiers in the QA gold sets as the retrieval relevance units.

The processed corpus summary contains 151,624 documents, 1,386,267 provisions, 1,513,376 chunks, 883,256 document relationships, and 19,763 external stubs. The relationship and validity fields require direction validation before they are used to make high-stakes legal filtering decisions.

B Additional Results

For the reranker family, candidate `Recall@30` and candidate `Hit@30` are 0.5995 and 0.6450 for every configuration. Gold retention@10 is 0.7129

for RRF-only, 0.8256 for mMiniLM, 0.8702 for Qwen3, 0.8630 for BGE, and 0.8850 for Jina. The source-derived Markdown report additionally records paired win/tie/loss counts, all requested slices, lexical safety diagnostics, and the status of the blinded annotation template. These values are retained in the machine-readable JSON rather than duplicated as a long appendix table.

C Run Manifests and Artifacts

All paths below are relative to the repository root. The lean commit retains manifests, aggregate metrics, reports, and latency summaries. Raw per-question outputs, retrieval caches, and the pending annotation template remain local evaluation artifacts and are intentionally not versioned here.

- Embedding text: [ablation_work/chunkOnly_vs_chunkMetadata/](#)
- Dense/hybrid/graph: [ablation_work/denseOnly_vs_denseGraph/](#) and [ablation_work/hybridOnly_vs_hybridGraph/](#)
- Reranker source and derived analysis: [ablation_work/reranker_choices/evaluation_runs/ablation3_reranker_v2/](#)
- Reranker analysis script: [ablation_work/reranker_choices/analyze_reranker_metrics.py](#)
- Generator-reasoning source runs and derived report: [ablation_work/e2e_LLM_Reasoning/](#) and [ablation_study/e2e_LLM_reasoning/results/](#)
- Generator-reasoning analysis script: [ablation_work/e2e_LLM_Reasoning/analyze_posthoc.py](#)
- Human-evaluation scaffold: generated locally for the pending semantic-evaluation phase and omitted from the lean commit.

The formal report’s Markdown counterparts are organized under [ablation_study/](#). The derived reports intentionally distinguish completed deterministic analyses from unavailable semantic legal-judge metrics and unmeasured full-online deployment statistics.

Table 9: Effect of graph expansion on top-10 gold evidence. “Graph-only” is gold evidence introduced by the graph context; “graph-missed” is gold evidence present before expansion but absent afterward. Counts are over 400 answerable questions.

Base	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Graph-only	Graph-missed
DenseOnly $\rightarrow$ DenseGraph	−0.1240	−0.0222	−0.0439	1 query (0.25%)	61 queries (15.25%)
HybridOnly $\rightarrow$ HybridGraph	−0.1181	−0.0240	−0.0411	6 queries (1.50%)	59 queries (14.75%)

D Retrieval and Reranker Configurations

Table 10 summarizes component settings. Family-specific budgets and generation settings appear in Section 5.3; optional system capabilities are distinguished from the components activated in each experiment.

Table 10: System-wide component and model configuration reference.

Component	Model / Checkpoint	Key Settings / Parameters	Role & Conventions
Dense Retriever	intfloat/multilingual-e5-large	560M params, 1024-dim, L2-normalized IP	Encodes Metadata-Enriched text (<code>Doc_Title Header Passage</code>); indexed in FAISS <code>IndexFlatIP</code> .
Sparse Retriever	BM25 (<code>rank_bm25</code>)	$k_1 = 1.5$, $b = 0.75$, <code>underthesea</code>	Retrieval/graph uses 10 independent shards with shard-local statistics; shard candidates are merged by BM25 score.
Fusion Layer	Reciprocal Rank Fusion (RRF)	$k = 60$, Top- $K = 30$ candidate pool	Fuses Dense and BM25 ranked hit lists into a shared candidate pool (C_{30}) for reranking.
Graph Module	G-LRAG (In-Memory)	Structural expansion; optional overlay	Retrieval/graph expands five seeds within their provisions, capped at 10 chunks. Query-time validity overlays are disabled in this family.
Reranker: mMiniLM	mmarco-mMiniLMv2-L12-H384-v1	117M params, FP32	Lightweight cross-encoder baseline; joint query-passage scoring over candidate pool C_{30} .
Reranker: BGE	BAAI/bge-reranker-v2-m3	567M params, Sigmoid activation	Multilingual cross-encoder; scores candidate passages; highest observed lexical answer overlap.
Reranker: Jina	jina-reranker-v2-base-multilingual	278M params, Position-ID patch	Multilingual reranker with position embedding compatibility patch; top overall retrieval quality.
Reranker: Qwen3	Qwen/Qwen3-Reranker-0.6B	0.6B params, FP16 precision	CausalLM-based reranker wrapper with automatic batch-size reduction and OOM retry logic.
Generator	gpt-4o-mini	$T = 0.0$, up to 10 context chunks	Retrieval/graph uses a Vietnamese evidence-only prompt and fallback phrase, with no explicit client-side output-token cap. Reasoning runs use a 1,024-token cap.